



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
ICS 3111: MICROPROCESSORS
END OF SEMESTER EXAM

Date: 1st August 2022

Time: 2 Hours

Instructions:

This Examination consists of **FOUR** questions

Answer **Question ONE (COMPULSORY)** and any other **TWO** questions.

Question one [20 Marks]

- What is the difference between IN and OUT instructions of the Intel 8085 2marks
- Write an 8085 assembly program to illustrate the difference between SHIFT and ROTATE instructions 4marks
- What are the control signals of DMA? 2marks
- An Intel 8085 programmer wants to implement a subroutine. Write a step by step procedure of this can be done successfully. 4marks
- Describe the steps involved in an opcode fetch operation of the Intel 8085 4marks
- The following code section is written an a programmer of Intel 8085 the T-state of each instruction is indicated calculate the total time it would take execute the code sections

Label	Opcode	Operand	Comments	T-state:
	LXI	B,2384H	;Load BC with 16-bit count	10
LOOP:	DCX	B	;Decrement (BC) by one	6
	MOV	A,C	;Place contents of C in A	4
	ORA	B	;OR (B) with (C) to set Zero flag	4
	JNZ	LOOP	;If result $\neq$ 0, jump back to LOOP	10/7

4marks

Question two [20 marks]

- a. Given the address bus structure of the Intel 8085. Describe how the bus structure determines the memory size of the microprocessor. 4marks
- b. With the aid of a diagram describe the function of each flag in the flag register of the Intel 8085 Microprocessor 4marks
- c. Use an appropriate diagram to describe how the ALE signal of the Intel 8085 functions 8marks
- d. List any TWO sixteen bit registers of the Intel 8085 microprocessor 2marks
- e. Two 8Kx8 EPROMs are to be interfaced with the Intel 8085 where address lines A_{15} - A_{13} are used for chip select through 3-to-8 decoder. Design the memory interfacing required in this case. 4marks

Question three [20 Marks]

- a. Write an assembly language program to multiply two numbers using the Intel 8085 assembly codes 6marks
- b. An assembly language programmer has been instructed to interface several sensors to the microprocessor using the 8255A general purpose programmable device. Write a control word and an 8085 assembly code that would make port A in input port and port B as an output. The port operation must rely on an external interrupt to function. 6marks
- c. List any four externally initiate control signals or interrupts of the Intel 8085 microprocessor 4marks
- d. List any FOUR functional grouping of the Intel 8085 instruction set. 4marks

Question four [20 Marks]

- a. With the aid of an example EACH, illustration ANY TWO categories of how Intel 8085 instructions are grouped in terms of the number of bytes. 2marks
- b. With the aid of a diagram illustrate show how the SIM register can be used to generate a control word where RST 7.5 is masked while RST 6.5 and 5.5 are unmasked. 6marks
- c. Write a program which will store the contents of accumulator and flag register at 2000 H and 2001 H memory locations respectively. 6marks
- d. Describe what is meant by peripheral mapped I/O scheme of the Intel 8085 2marks

- e. 2 KB RAM, 2 KB ROM, one input and one output device are to be interfaced with 8085 microprocessor. Employ memory mapped I/O scheme to execute the above. 4marks